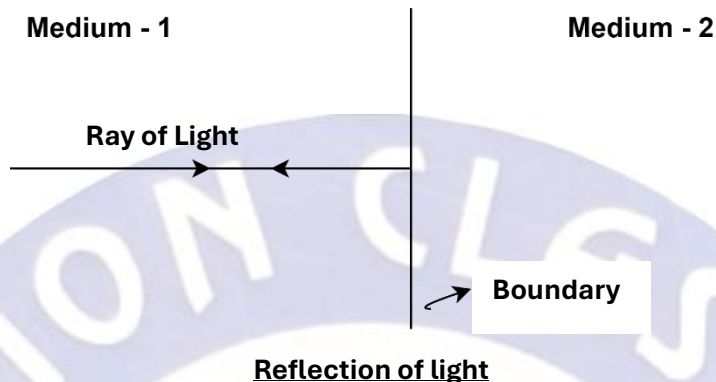


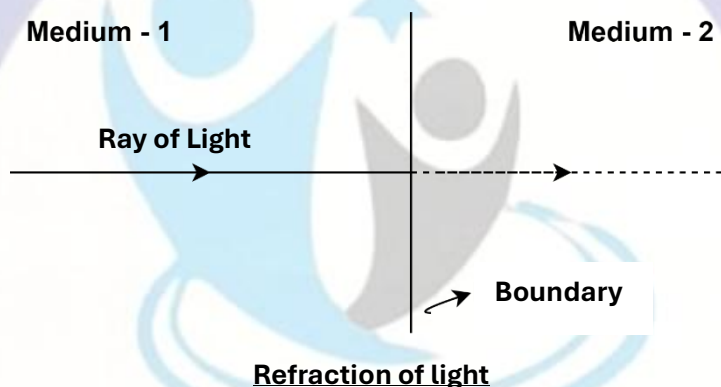
## REFRACTION OF LIGHT

- Introduction

Case 1:

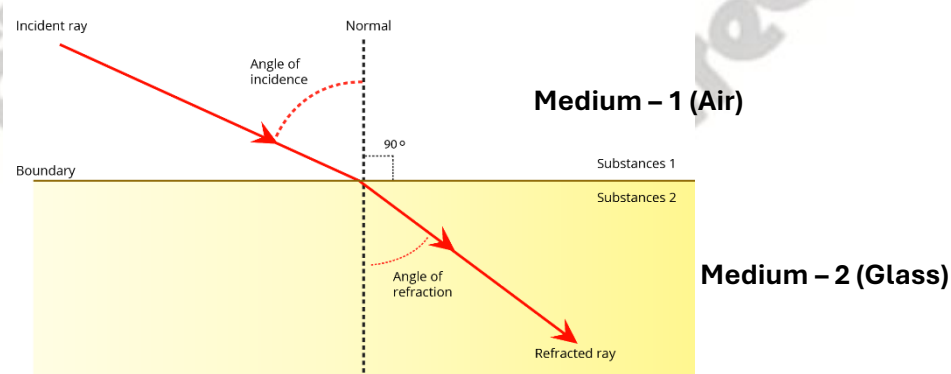


Case 2:



- Refraction of light

- When the ray of light passes through one medium to another medium then it deviates either towards the normal or away from the normal, the phenomenon of bending of light is called Refraction of light.



- Refraction of light states as two laws:-

- The incident ray, the refracted ray and normal at the point of incidence are lie on the same plane.
- For a particular medium, the ratio of the sine of angle of incidence ( $\sin i$ ) to the of the angle of refraction ( $\sin r$ ) is constant.

i.e,  $\frac{\sin i}{\sin r} = \text{Constant}$

→ This constant is also called snell's constant.

This law is also known as Snell's law.

**Note:**

- This constant ratio of  $\left(\frac{\sin i}{\sin r}\right)$  is called the refractive index of Medium – 2 with respect to (w.r.t) Medium – 1.  
 i.e, Refractive index of Medium – 2 w.r.t Medium – 1 =  $\frac{\sin i}{\sin r}$

• **Refractive index**

- The ratio of speed of light in vacuum and speed of light in any medium is called refractive index of that medium.
- It is denoted by 'μ' or 'n'.  
 ↳ Mew

i.e, Refractive index of medium =  $\frac{\text{Speed of light in vacuum}}{\text{Speed of light in medium}}$

If the speed of light in vacuum =  $C = 3 \times 10^8 \text{ m/s}$

&

Speed of light in any medium =  $V$

∴  $n = \frac{C}{V}$

OR

$V = \frac{C}{n}$

- The S.I unit of Refractive index =  $\frac{\text{S.I unit of } C}{\text{S.I unit of } V}$   
 $= \frac{\text{m/s}}{\text{m/s}} = 1$

**Note:**

- Refractive index  $\propto$  Density of medium

Thus, Refractive index is a unitless quantity.

- **Thus, Refractive index of a medium can be obtain by three ways:**

Case 1: In terms of angle [i & r]

Refractive index =  $\frac{\sin i}{\sin r} = \mu_{21} = n_{21}$

Case 2: In the terms of refractive index

i.e,  $n_{21} = \frac{n_2}{n_1}$

Read as

→ Refractive index of a Medium – 2 w.r.to Medium – 1

- $n_{21}$  is also written as  $\mu_{21}/\mu_2$

**Case 3: In the terms of velocity**

- Refractive index of Medium – 2 w.r.to Medium – 1 is denoted by  $n_{21}/\mu_{21}/{}_1\mu_1$ .

$$\text{Now, } n_{21} = \frac{n_2}{n_1} = \frac{\frac{c}{v_2}}{\frac{c}{v_1}} = \frac{v_1}{v_2}$$

Where,

$V_1$  = Velocity of Light in Medium – 1.

&

$V_2$  = Velocity of Light in Medium – 2.

$$\Rightarrow \boxed{n_{21} = \frac{v_1}{v_2}}$$

• **Relative Refractive index**

- When light is going from one (other than vacuum or air) to another medium, then the value of refractive index is called relative refractive index.

**For example:-**

- ✓ The refractive index of glass with respect to water is 1.125.

• **Absolute Refractive index**

- When light is going from vacuum or air to another medium, then the value of refractive index is called absolute refractive index.

OR

The refractive index of a medium with respect to air is also called its absolute refractive index.

**For example:-**

- ✓ Refractive index of water with respect to air is 1.33.

**For Example-**

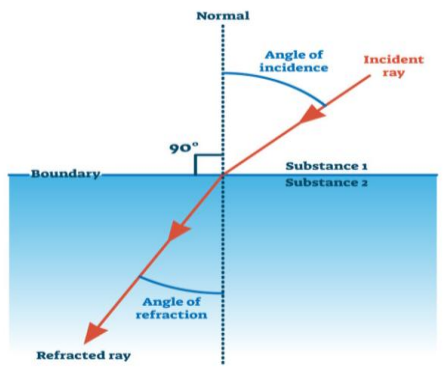
| Medium               | Refractive Index |
|----------------------|------------------|
| Air                  | 1.0003           |
| Ice                  | 1.31             |
| Water                | 1.33             |
| Alcohol              | 1.36             |
| Ethyl alcohol        | 1.36             |
| Kerosene             | 1.44             |
| Fused Quartz (glass) | 1.46             |
| Benzene              | 1.50             |
| Crown Glass          | 1.52             |
| Carbon disulphide    | 1.63             |
| Flint Glass          | 1.65             |
| Flint Glass          | 1.66             |
| Diamond              | 2.42 (↑)         |

Q. The refractive index of diamond is 2.42. What is the meaning of this statement?

Ans The refractive index of diamond is 2.42. It means that the speed of light in diamond is  $(1/2.42)^{\text{th}}$  part of the speed of light in vacuum.

▪ **Effect of refraction of Light**

1. A ray of light bends or refracts as it enters a different medium.
2. Refraction of light is responsible for the bent image of pen, pencil, spoon, rod etc.



3. Enlarge image of coin below the water/Liquid.
4. Enlarge size of rasgulla below the water (sugar juice).
5. Pool of water appears to be less deep than what it actually is.
6. The volume of bucket full of water appears less than actual.
7. The depth of fish/coin/rasgulla deeped in water appears above than actual depth.
8. The twinkling of stars.

**Special Note:**

- In the process of refraction of light frequency is same but wavelength, speed, amplitude & intensity decreases.

• **Numerical Based Problem**

Q1. If light ray enters from air to a glass plate having refractive index is 1.5. What is the speed of light in glass.

Ans Since Given,

Refractive index of glass ( $\mu$ ) = 1.5

$$\therefore n = \frac{c}{v} \Rightarrow v = \frac{c}{n}$$

$$\Rightarrow v = \frac{3 \times 10^8 \text{ m/s}}{1.5} = 2 \times 10^8 \text{ m/s}$$

$\therefore$  Speed of light in glass =  $2 \times 10^8 \text{ m/s}$ .

Q2. Find the value of  ${}_1\mu_2 \times {}_2\mu_3 \times {}_3\mu_1$ .

Ans  $\therefore \frac{{}_1\mu_2 \times {}_2\mu_3 \times {}_3\mu_1}{\mu_1 \times \mu_2 \times \mu_3} = 1.$

Q3. A ray of light travelling in air and falls on a surface of transparent slab. If the angle of incidence is 45. find the angle of refraction if refractive index of glass slab is  $\sqrt{2}$ .

Ans Since given,

$n =$  refractive index of glass slab =  $\sqrt{2}$ .

angle of incidence ( $i$ ) = 45.

$$\therefore n = \frac{\sin i}{\sin r}$$

$$\begin{aligned} \Rightarrow \sqrt{2} &= \frac{\sin 45^\circ}{\sin r} \Rightarrow \sqrt{2} = \frac{\frac{1}{\sqrt{2}}}{\sin r} \\ \Rightarrow \sqrt{2} \times \sin r &= \frac{1}{\sqrt{2}} \\ \Rightarrow \sin r &= \frac{1}{\sqrt{2} \times \sqrt{2}} = \frac{1}{2} \\ \Rightarrow \sin r &= \sin 30^\circ \Rightarrow r = 30^\circ \\ \therefore \text{Angle of refraction} &= 30^\circ \end{aligned}$$

**Q4.** The light ray enters from air into a glass plate having refractive index 1.5. What is the speed of light in the glass.

**Ans** Since given,

refractive index (n) of glass = 1.5

&  $C = 3 \times 10^8 \text{ m/s}$

$$\therefore n = \frac{C}{V} \Rightarrow V = \frac{C}{n}$$

**Q5.** If the refractive index of glass w.r.to air is  $\frac{3}{2}$  and refractive index of water w.r.to air is  $\frac{4}{3}$ . Find the refractive index of glass w.r.to water?

**Ans** Since given,

$$n_{GA} = \frac{3}{2} \Rightarrow \frac{n_G}{n_A} = \frac{3}{2} \dots\dots\dots(i)$$

&

$$n_{WA} = \frac{4}{3} \Rightarrow \frac{n_W}{n_A} = \frac{4}{3} \dots\dots\dots(ii)$$

$$\therefore n_{GW} = ?$$

On dividing equation (i) by equation (ii)

$$\begin{aligned} \Rightarrow \frac{\frac{n_G}{n_A}}{\frac{n_W}{n_A}} &= \frac{\frac{3}{2}}{\frac{4}{3}} = \frac{3 \times 3}{2 \times 4} \\ \Rightarrow n_{GW} &= \frac{9}{8} = 1.125 \end{aligned}$$

$\therefore$  Refractive index of glass w.r.to air = 1.125

**Q6.** A ray of light enters a diamond from air. If the refractive index of diamond is 2.42 by what percent does the speed of light reduce on entering the diamond.

**Ans** Since Given,

Refractive index of diamond (n) = 2.42

$$\begin{aligned} \therefore \text{we know that, } n &= \frac{C}{V} \\ \Rightarrow V &= \frac{C}{n} = \frac{1C}{2.42} = \frac{100}{242} \times C \\ \Rightarrow V &= 0.41 C \end{aligned}$$

$$\therefore \text{Reduced light} = 1 C - 0.41 C = 0.59$$

$$\begin{aligned} \therefore \text{Reduced light \%} &= 0.59 \times 100 \\ &= 59 \% \end{aligned}$$

The light reduced in diamond is 59 %.

**Q7.** If the refractive index of water w.r.to air is  $\frac{4}{3}$  then find the refractive index of air w.r.to water.

$$\text{Ans } \therefore n_{WA} = \frac{4}{3} \Rightarrow \frac{n_W}{n_A} = \frac{4}{3}$$

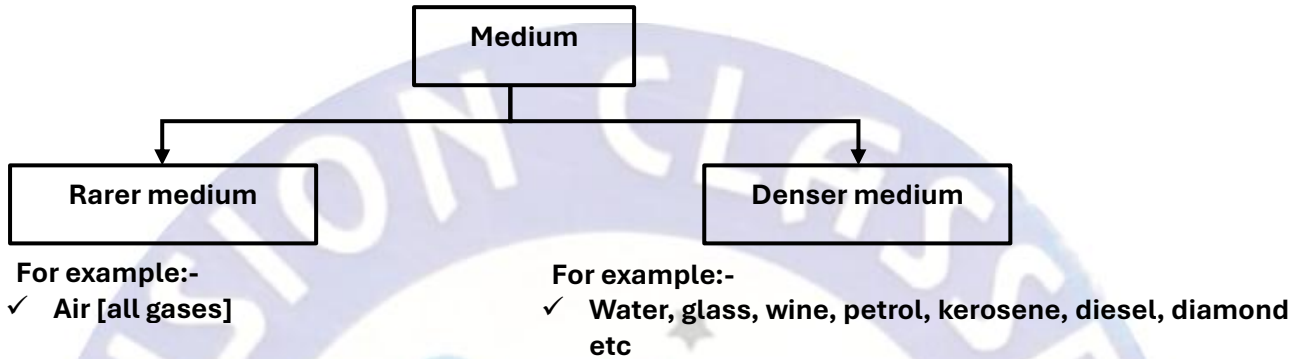
$$\Rightarrow \frac{n_A}{n_W} = \frac{3}{4} \quad \Rightarrow n_{AW} = \frac{3}{4}$$

Q8. If the refractive index of glass w.r.to air is  $\frac{3}{2}$ . Find the refractive index of air w.r.to glass.

Ans  $\because n_{GA} = \frac{3}{2} \quad \therefore n_{AG} = \frac{2}{3}$

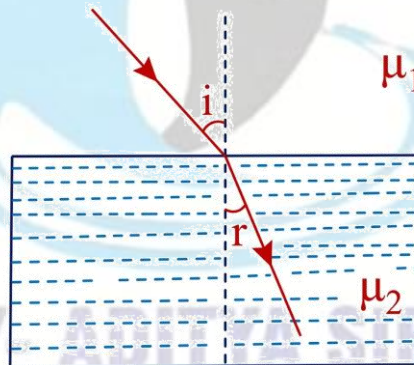
- **Medium**

➤ Where the light can travel, called medium.

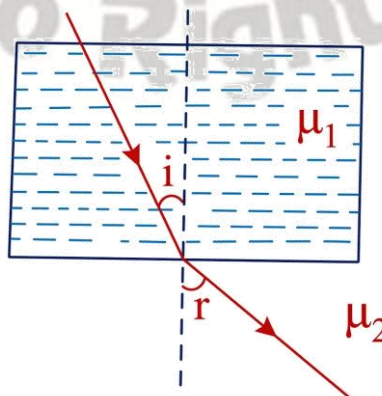


**Note:**

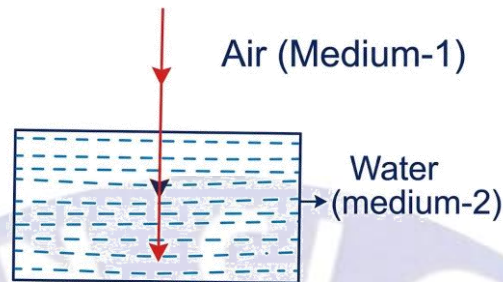
1. If the ray of light enters from rarer medium to denser medium then ray bends towards the normal.  
i.e,  $i > r$  So  $\mu_2 > \mu_1$



2. If the ray of light enters from denser medium to rarer medium then ray bends away from the normal.  
i.e,  $\mu_2 < \mu_1$  /  $n_2 < n_1$  OR  $i < r$



3. If the ray of light falls perpendicular to the medium then it passes through the medium without any deviation.



4. Refractive index depends on the nature of material of the medium and on the wavelength of the light.

- **Angle of Emergence**

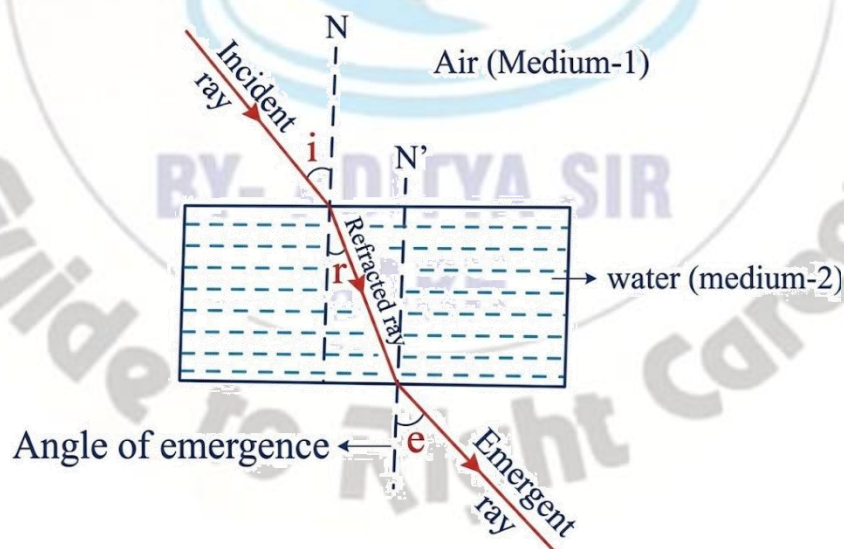
- The angle between emergent ray and normal is called angle of emergence.
- It is denoted by 'e'.

- **Emergent ray**

- A ray which originates from a medium after refraction and exists glass slab parallel to the incident ray of light is called emergent ray.

OR

When a ray of light enters from air to any medium, after refraction it outside from the medium which is parallel to the incident ray is called emergent ray.

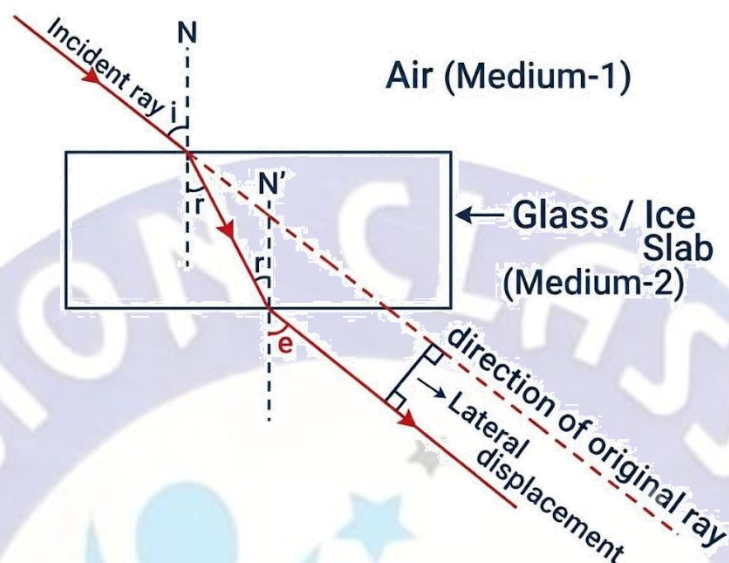


**Note:** Angle of emergence depends

- (i)  $e \propto i$
- (ii)  $e \propto \text{refractive index of medium}$
- (iii)  $e \propto \text{thickness of medium}$
- (iv)  $e \propto \frac{1}{\lambda}$   
                     └─→ wavelength

- **Lateral Displacement**

- A ray of light travels from air to any medium, after refraction through a glass slab, it emerges in a direction parallel to the original direction but slightly displaced from the line i.e, the perpendicular distance between emergent ray and original path of incident is called lateral displacement.



- **Factors of Lateral Displacement**

1. Lateral Displacement  $\propto$  Angle of incidence ( $i$ ).
2. Lateral Displacement  $\propto$  thickness of slab.
3. Lateral Displacement  $\propto$  refractive index of glass slab ( $n$ ).
4. Lateral Displacement  $\propto \frac{1}{\text{wavelength of light}}$

- **Critical Angle**

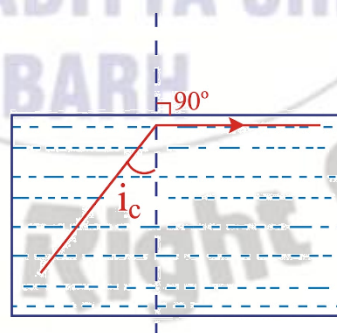
- When a ray of light travels from denser to rarer medium i.e, the angle of incidence is directly proportional to the angle of refraction.

For a particular value of angle of incidence ( $i$ ) is called critical angle, the angle of refraction becomes  $90^\circ$ .

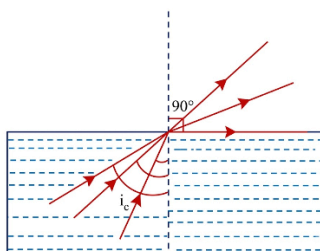
- It is denoted by ' $i_c$ '.

$$\text{Thus, } \frac{n_2}{n_1} = \frac{\sin i}{\sin r} = \frac{\sin i_c}{\sin 90^\circ} = \frac{\sin i_c}{1}$$

$$\Rightarrow \sin i_c = \frac{n_2}{n_1}$$

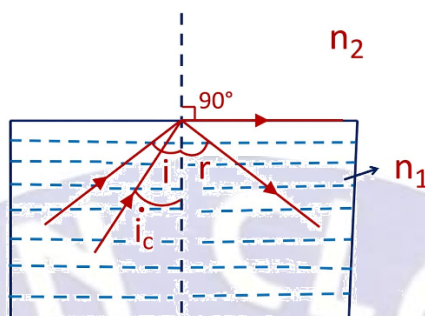


**Special Fundamental**



- **Total internal reflection**

- The total internal reflection of light travelling in a medium of higher refractive index when it incidents on the boundary with the medium of lower refractive index at an angle greater than the critical angle, is called total internal reflection.



**Note:**

- When angle of incidence is greater than critical angle i.e, ray returns back into same medium (Medium – 1) So, refraction is not possible.

- **Refracted ray**

- When a ray of light travels from one medium to the other medium then it deviates this deviated ray is called refracted ray.

- **Angle of refraction**

- The angle between refracted ray and normal is called angle of refraction.
- It is denoted by 'r'.

- **Lens**

- A transparent material bounded by two surfaces of which at least one surface is curved called lens.

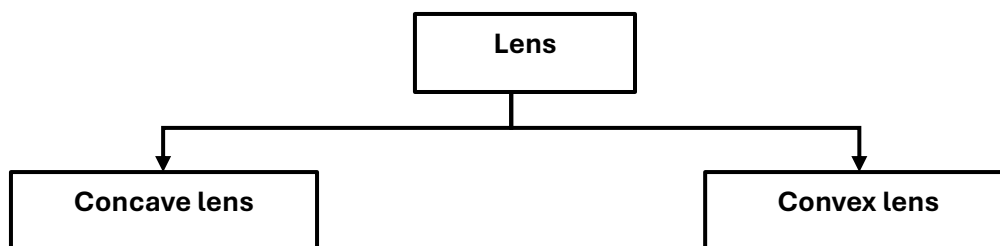
OR

A lens is a piece of transparent glass bounded by two spherical surfaces.

- **Lens is an Optical Device**

**Uses of lens**

1. In the Spectacles.
2. In the magnifying glass.
3. In Cameras.
4. In the microscopes.
5. In the projectors.



### Uses of Concave lens

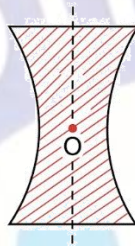
1. Correcting in Myopia.
2. Galileo microscope.

### Uses of Convex lens

1. In compound microscope.
2. Correcting in Hypermetropia.
3. Telescope.

- **Concave lens**

- That lens which is thicker at the edges and thinner at the middle is called Concave lens.

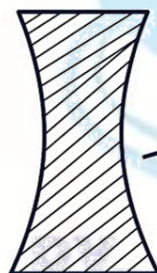


Concave lens

### Types of Concave lens

1. **Bi – Concave lens**

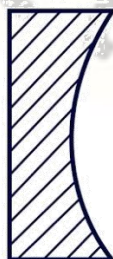
- That concave lens which both surface are concave, called bi – concave lens.



→ Biconcave

2. **Plano - Concave lens**

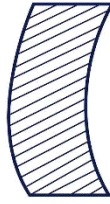
- That concave lens which one surface is plane and another surface is concave, called Plano – concave lens.



plano-concave lens.

3. **Convexo - Concave lens**

- That lens which one surface is convex and other surface is concave, called convexo – concave lens.

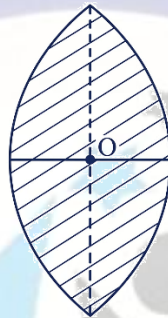


- **Aperture**

- The diameter of the circular perimeter of a lens is called aperture.

- **Convex lens**

- A lens that is thicker at the middle than at the edge is called convex lens.



**Types of Convex lens**

1. **Bi – Convex lens**

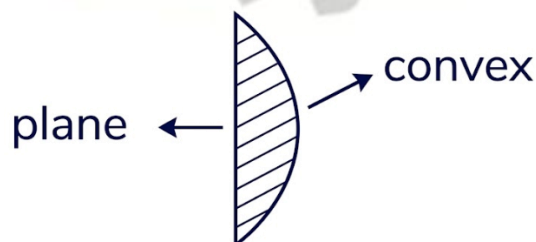
- That lens which both surfaces are convex is called bi – convex lens.



**Bi-convex lens**

2. **Plano - Convex lens**

- That lens which one surface is plane and other surface is convex, called plano - convex lens.



### 3. Concavo - Convex lens

- That lens which one surface is concave and other surface convex, called concavo – convex lens.



concavo-convex lens

#### **Note:**

- These kinds of lenses can be made by very thin, such lenses are called thin lenses.

#### ▪ Some important terms related to thin lenses

##### 1. Optical Centre

- The central point of the lens is called optical centre. It is denoted by 'O'.

##### 2. Principal axis

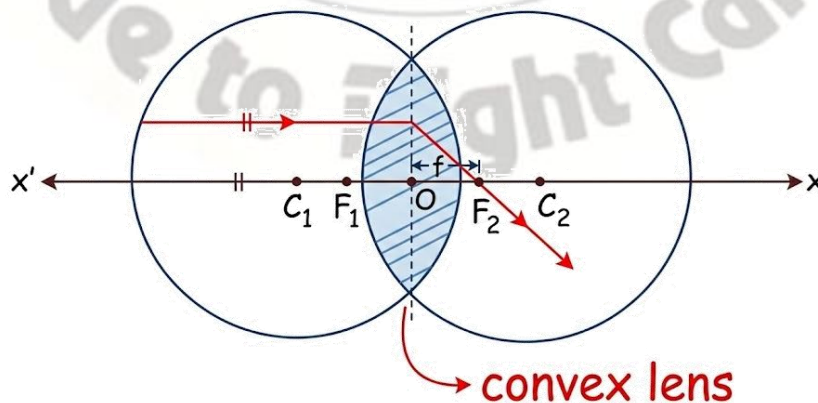
- A lens is bounded by two spherical surface and therefore it has two centre of curvature  $C_1$ , &  $C_2$ . The line joining  $C_1$  &  $C_2$  is called principal axis.
- It is denoted by  $XX'$

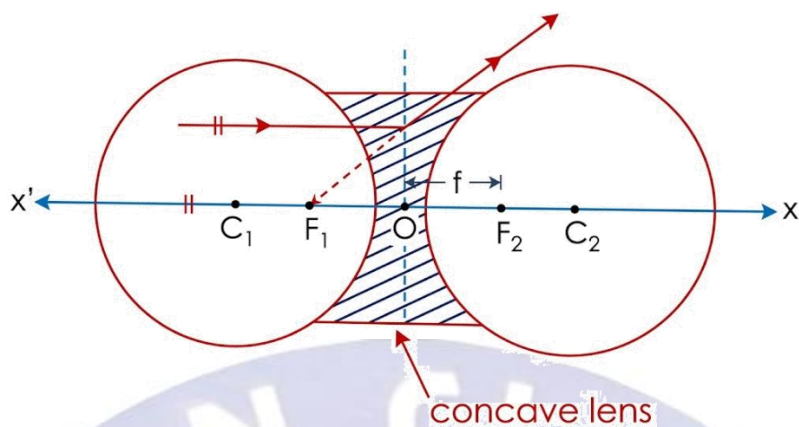
##### 3. Focus of a lens

- If a beam of light incidents parallel to the principal axis of a convex lens, It converges to a point  $F_2$  on the principal axis on the other side of the lens. This point is called principal focus of lens.
- It is denoted by  $F_2$  (F).

##### 4. Focal length

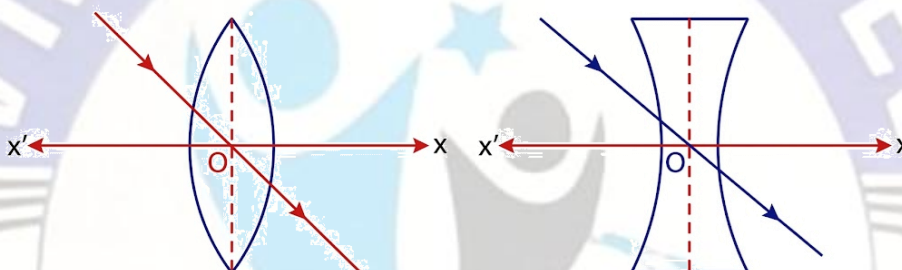
- The distance of the focus from the optical centre, is called the focal length.
- It is denoted by 'f'.



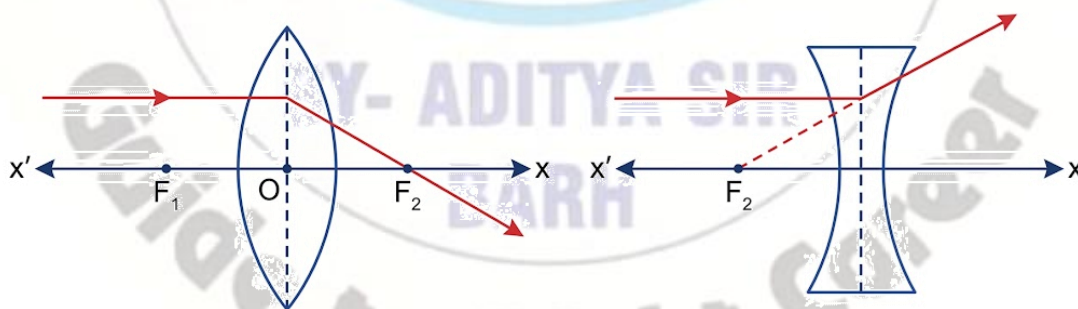


▪ **Ray diagram in lens**

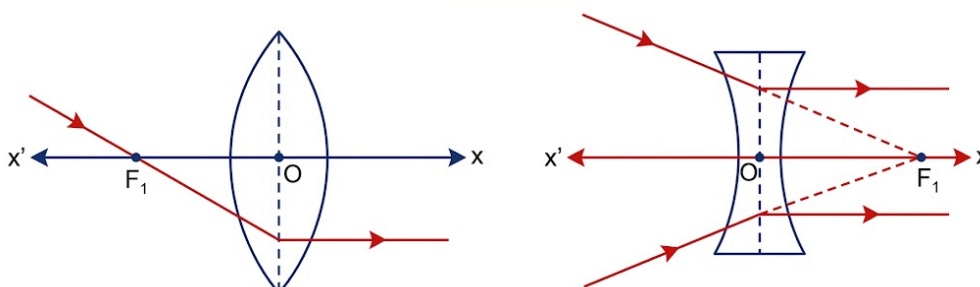
**Case (i)** When a ray of light incident towards the optical centre (O) passes through without any deviation.



**Case (ii)** When the ray of light incidents parallel to the principal axis, after refraction it either passes or appear to pass through focus.



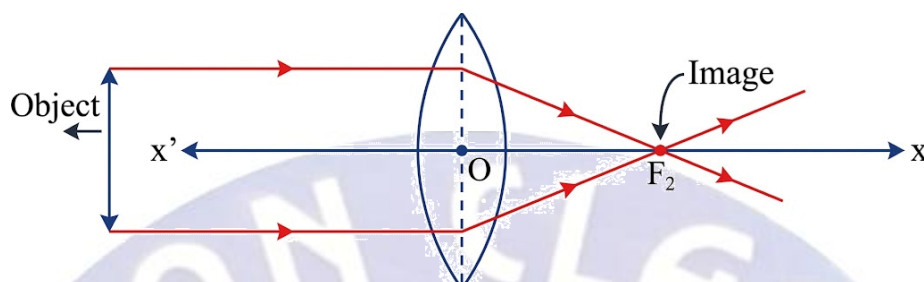
**Case (iii)** When a ray of light passes through focus, after refraction it always passes or appear to pass through parallel to the principal axis XX'.



▪ **Image formation in lens**

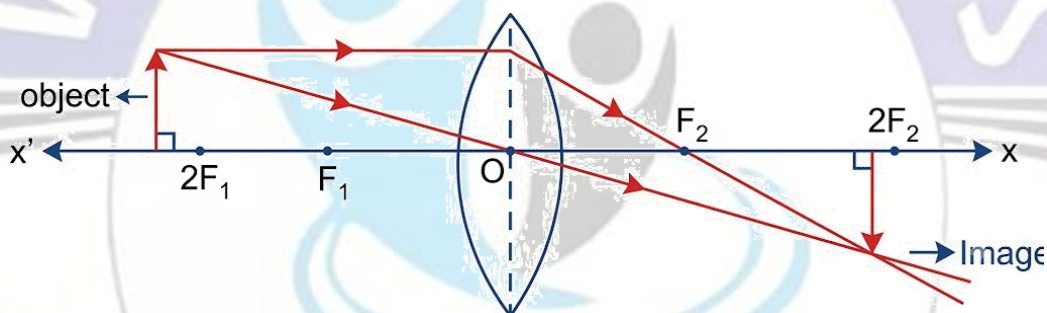
**Case A: In the Convex lens**

**Case (i) When an object placed at infinity**



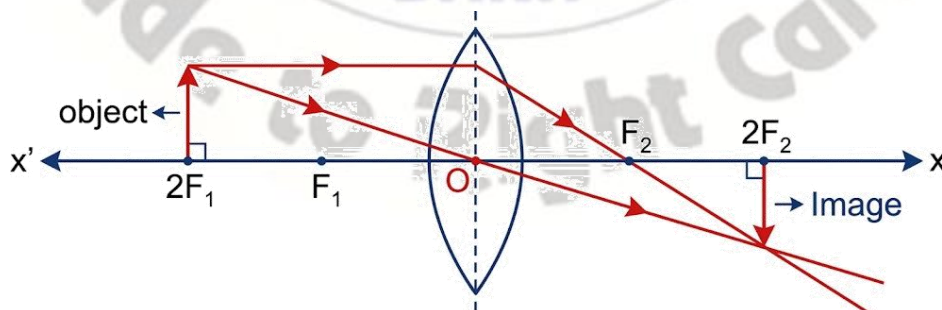
1. Position of image = at the focus ( $F_2$ ).
2. Nature of image = real and inverted.
3. Size of image = much smaller than object i.e, point sizes.

**Case (ii) When an object placed between infinity and  $2F_1$ .**



1. Position of image = between  $F_2$  &  $2F_2$ .
2. Nature of image = real and inverted.
3. Size of image = Smaller than object.

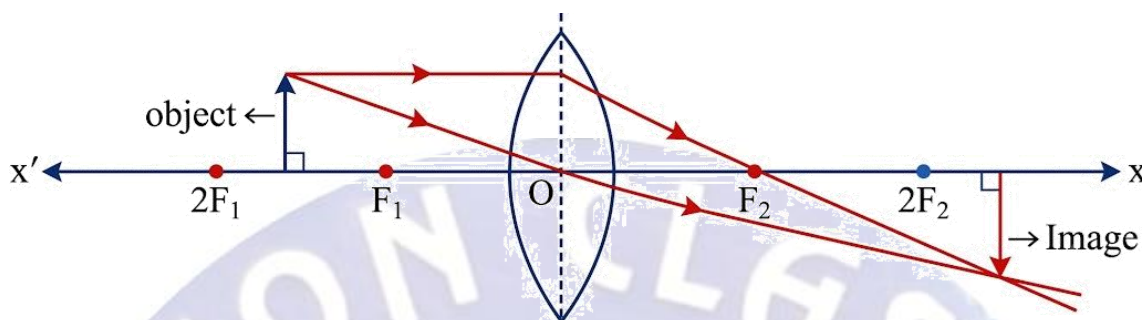
**Case (iii) When an object is placed at  $2F_1$ .**



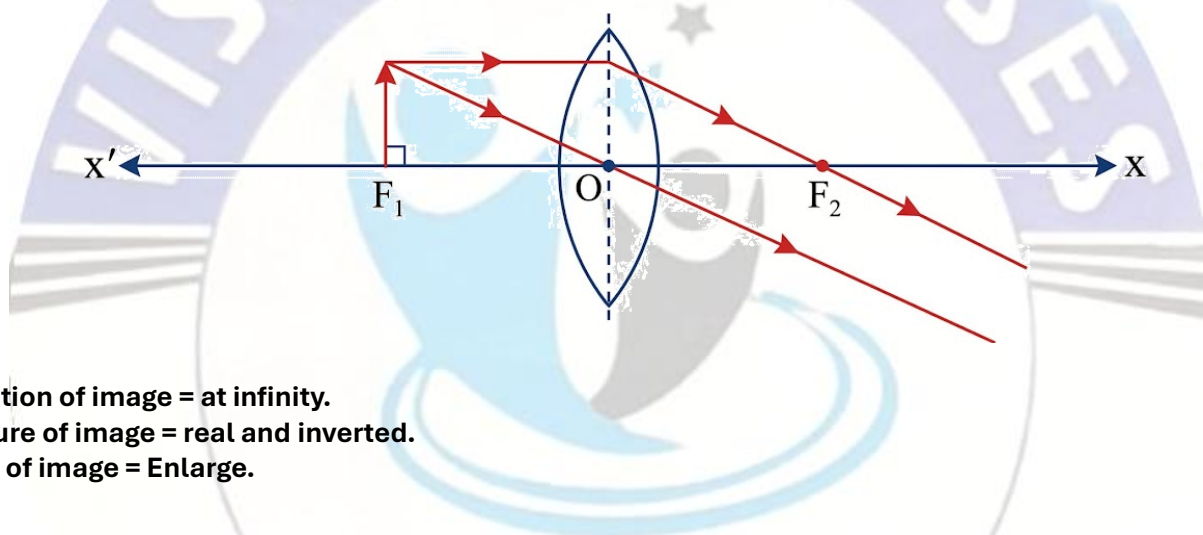
1. Position of image = at  $2F_2$ .
2. Nature of image = real and inverted.
3. Size of image = same size as object.

**Case (iv) When an object is placed between  $F_1$  and  $2F_1$ .**

1. Position of image = beyond  $2F_2$ .
2. Nature of image = real and inverted.
3. Size of image = enlarged.

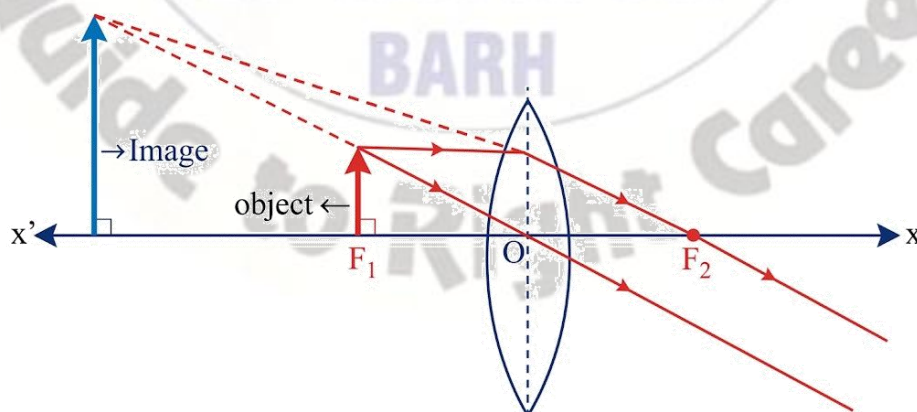


**Case (v) When an object is placed at Focus ( $F_1$ ).**



1. Position of image = at infinity.
2. Nature of image = real and inverted.
3. Size of image = Enlarge.

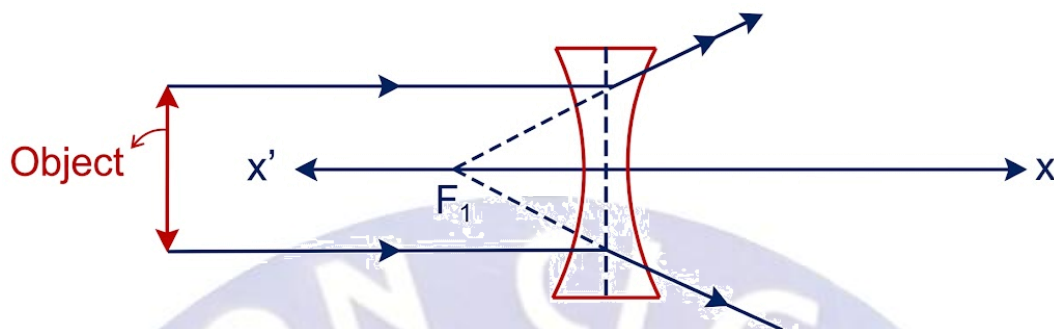
**Case (vi) When an object is placed between optical centre O and focus (F).**



1. Position of image = inside the object / On the same side as object.
2. Nature of image = virtual and erect.
3. Size of image = Larger than object.

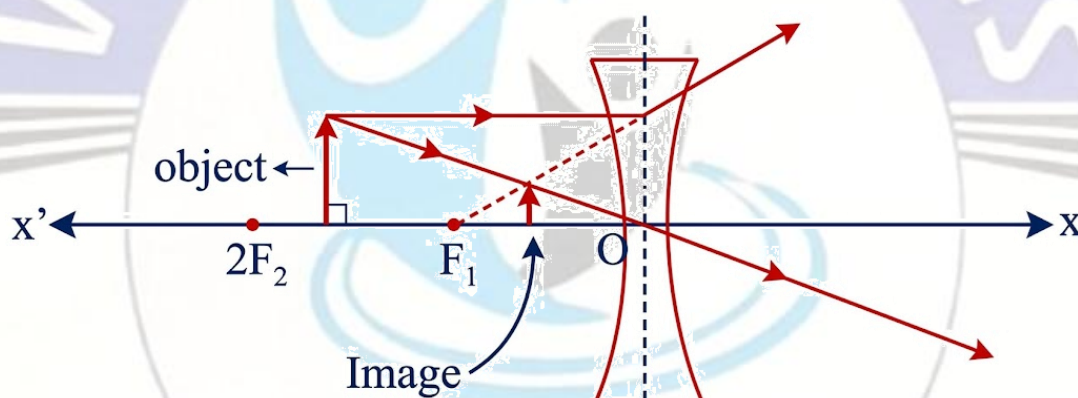
### Case B: In Concave lens

Case (i) When an object is placed at infinity.



1. Position of image = at focus  $F_1$ .
2. Nature of image = Virtual & erect.
3. Size of image = diminished.

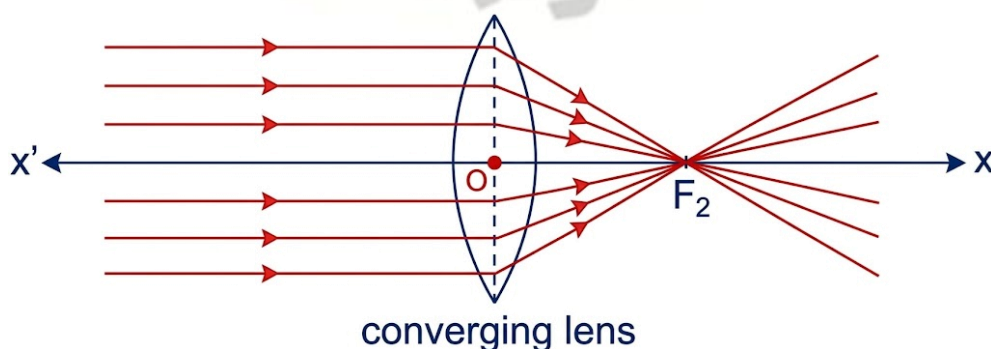
Case (ii) When an object placed between  $F_1$  and  $2F_2$ .



1. Position of image = between O and  $F_1$ .
2. Nature of image = virtual and erect.
3. Size of image = smaller than object.

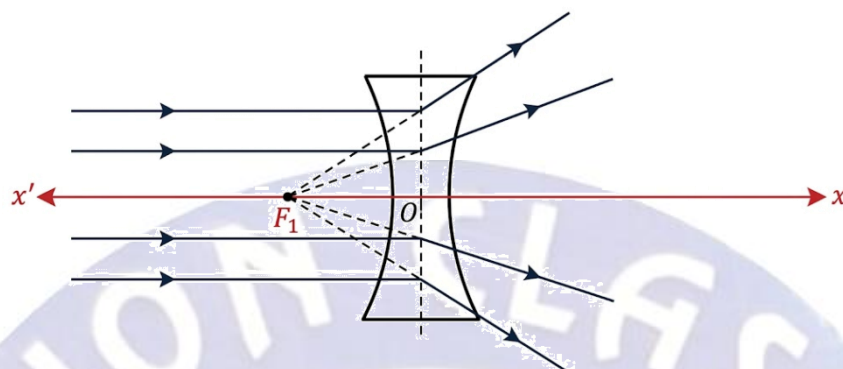
Q. Convex lens is also called Converging lens why? Explain it !

Ans When the rays of light travels parallel to the principal axis in a convex lens, after refraction they converge at a certain point on the principal axis of another side of lens, the converging property at convex lens is called converging lens.



**Q. Concave lens is also called diverging lens why? Explain it!**

**Ans** When the rays of light passing through parallel to the principal axis of a concave lens, after refraction they diverges the rays on another side and appear to converge inside the incident ray due to the diverging properties of a concave lens is also called diverging lens.



**Q. If you are given three lenses: Concave, Convex & a circular glass how will you identify them without touching them?**

**Ans** To identify the lenses, we will do an experiment, First we would take a pencil and put it one – by one in front of the lenses and observe the image, if the formed image is larger than pencil then it is convex, if formed image is smaller than it concave lens and formed image is same size as object then it is circular glass.

• **Power of a lens**

- The power of a lens is defined as “the reciprocal of its focal length” where unit of length is “meter”.
- The power of a lens is denoted by ‘P’.

Thus,  $\text{Power of a lens} = \frac{1}{\text{focal length}}$

i.e,  $P = \frac{1}{f}$ , [  $f$  is in metre]

**Note:**

$\therefore P \propto \frac{1}{f}$

If  $f$  is in centimetre (cm)

Then,  $P = \frac{1}{\left(\frac{f}{100}\right)} = \frac{100}{f}$

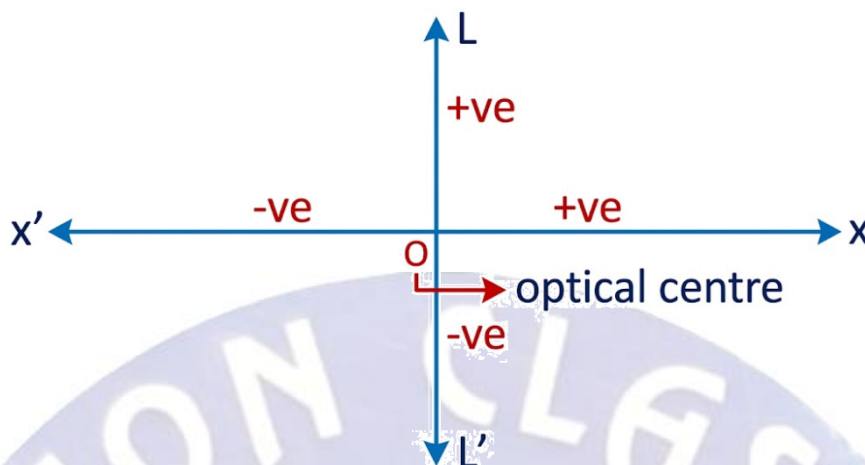
$\Rightarrow$   $P = \frac{100}{f}$

- The S.I unit of power of lens is  $\frac{1}{m} = 1 \text{ m}^{-1}$ .
- The famous unit/S.I unit of power of lens is ‘diopter’ and it is denoted by ‘D’.

**Note:**

1.  $1 \text{ m}^{-1} = 1 \text{ Diopter (1D)}$
2. The power of lens of convex lens is positive because focal length of convex lens is positive.  
i.e,  $p = +ve$  [ $f = +ve$ ]
3. The power of a concave lens is negative because the focal length of concave lens is negative.  
i.e,  $p = -ve$  [ $f = -ve$ ]

• **Sign Convention**



**Note:**

1. The focal length ( $f$ ) of convex lens is positive.
2. The focal length ( $f$ ) of concave lens is negative.

Q. If the focal length of concave lens is 20 cm. find its power of lens?

Ans  $\therefore$  focal length of concave lens ( $f$ ) = -20 cm.

&

$$P = \frac{100}{f}$$

$$\Rightarrow P = \frac{100}{-20} = -5 \text{ D}$$

$\therefore$  Power of concave lens = -5 D

Q. If the focal length of a convex lens is 0.25 m. find its power.

Ans  $\therefore$  focal length of convex lens ( $f$ ) = 0.25 m.

&

$$P = \frac{1}{f} = \frac{1}{0.25} = \frac{100}{25} = 4 \text{ D}$$

$\therefore$  Power of convex lens = 4 D

• **Combination of lenses or Contact of lenses**

➤ If two or more lenses are in contact then these are called combination/Contact of lens.

Case 1: If two lenses of focal lengths  $f_1$  and  $f_2$  are in contact.

Let, focal length of combined lenses is  $f$ .

$$\therefore \frac{1}{f} = \frac{1}{f_1} + \frac{1}{f_2} \quad \Rightarrow \frac{1}{f} = \frac{f_1 + f_2}{f_1 \times f_2}$$

Case 2: If more than two lenses of focal length  $f_1, f_2, f_3, \dots, f_n$  are in contact.

Let, Combined focal length =  $f$

$$\therefore \frac{1}{f} = \frac{1}{f_1} + \frac{1}{f_2} + \frac{1}{f_3} + \dots + \frac{1}{f_n}$$

• **Power of Combination of lenses**

- Let power of lenses are  $P_1, P_2, P_3, \dots, P_n$  and its combined power is  $P$ .  
 $\therefore P = P_1 + P_2 + P_3 + \dots + P_n$

**Note:**

- Combination of lenses is used in telescope, microscope, photographic camera etc.

**For Example:**

Q. If two lenses of focal length 20 cm and the 10 cm are in contact. Find its combined power.

Ans  $\therefore f_1 = 20 \text{ cm}$  &  $f_2 = 10 \text{ cm}$

$$\therefore f = \frac{f_1 \times f_2}{f_1 + f_2} = \frac{20 \times 10}{20 + 10}$$

$$\Rightarrow f = \frac{200}{30} = \frac{20}{3} \text{ cm}$$

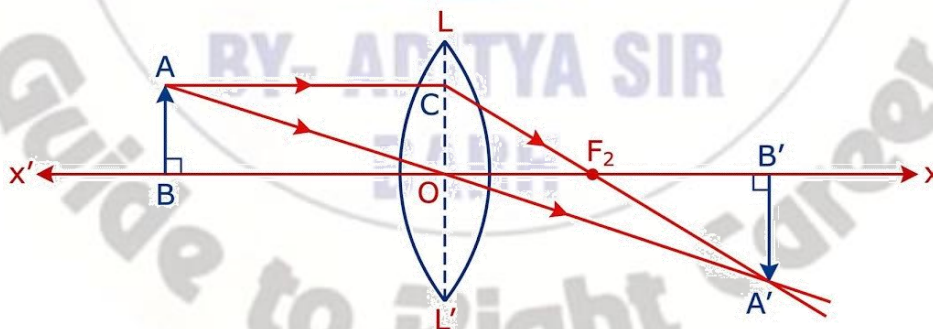
$$\therefore P = \frac{100}{f} \Rightarrow P = \frac{100}{\frac{20}{3}} = \frac{300}{20} = 15$$

$$\Rightarrow P = 15 \text{ D}$$

- **To obtain lens formula:**  $\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$

**Case 1: In Convex lens**

- Let  $LL'$  is a convex lens of focus  $F$ , Focal length of optical centre of principal axis  $XX'$ .  
 ➤ A luminous object  $AB$  is placed on the  $XX'$ .  
 ➤ A real image  $A'B'$  is formed opposite sides of lens and below the  $XX'$ .



In  $\triangle AOB$  and  $\triangle A'B'O$

$$\angle AOB = \angle A'OB' \quad [\text{V.O angle}]$$

$$\angle ABO = \angle A'B'O$$

$\therefore$  by A – A similarity

$$\triangle AOB \sim \triangle A'B'O$$

$$\frac{AB}{A'B'} = \frac{OB}{OB'} \quad \dots\dots (i)$$

Similarity, In  $\triangle COF_2$  and  $\triangle A'B'F_2$

$$\angle COF_2 = \angle A'B'F_2 = 90^\circ$$

$$\angle CF_2O = \angle A'F_2B$$

∴ by A – A similarity

$$\Delta COF_2 \sim \Delta A'B'F_2$$

$$\Rightarrow \frac{CO}{A'B'} = \frac{OF_2}{B'F_2}$$

∵ AC || XX'

∴ CO = AB (distance between two parallel lines)

$$\therefore \frac{AB}{A'B'} = \frac{OF_2}{B'F_2} \quad \dots\dots (ii)$$

From equation (i) and (ii)

$$\frac{OB}{OB'} = \frac{OF_2}{B'F_2} \Rightarrow \frac{OB}{OB'} = \frac{OF_2}{OB' - OF_2}$$

Using sign convention,

OB = distance of object = -u

OB' = distance of image = v

OF<sub>2</sub> = focal length = f

$$\therefore \frac{OB}{OB'} = \frac{OF_2}{OB' - OF_2} \Rightarrow \frac{-u}{v} = \frac{f}{v - f}$$

$$\Rightarrow vf = -u(v - f)$$

$$\Rightarrow vf = -uv + uf$$

Both sides divided by uvf

$$\frac{vf}{uvf} = \frac{-uv}{uvf} + \frac{uf}{uvf}$$

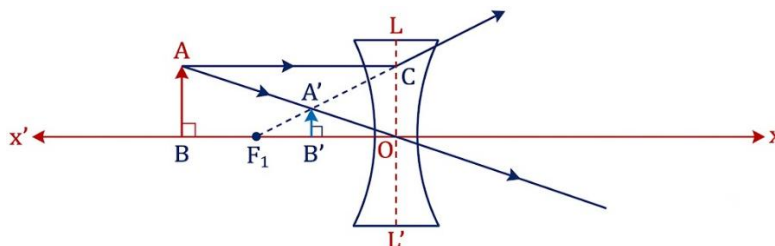
$$\Rightarrow \frac{1}{u} = \frac{-1}{f} + \frac{1}{v}$$

$$\Rightarrow \frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

→ Which is required lens formula.

### Case 2: In Concave lens

- Let LL' is a concave lens of optical centre O, focus F, focal length f and principal axis XX'.
- A luminous object AB is kept on the principal axis XX'.
- A virtual and erect image A'B' is formed towards to the lens.



Now, In  $\triangle ABO$  and  $\triangle A'B'O$   
 $\therefore \angle ABO = \angle A'B'O = 90^\circ$   
 &  $\angle AOB = \angle A'OB'$  [Common angle]

By A – A similarity,  
 $\triangle ABO \sim \triangle A'B'O$   
 $\therefore \frac{AB}{A'B'} = \frac{OB}{OB'} \dots\dots\dots (i)$

Similarly, In  $\triangle COF_1$  and  $\triangle A'B'F_1$   
 $\angle COF_1 = \angle A'B'F_1 = 90^\circ$   
 &  $\angle CF_1O = \angle A'F_1B'$  [Common angle]

By A – A similarity,  
 $\triangle COF_1 \sim \triangle A'B'F_1$   
 $\frac{CO}{A'B'} = \frac{OF_1}{B'F_1}$   
 $\Rightarrow \frac{AB}{A'B'} = \frac{OF_1}{B'F_1} \dots\dots\dots (ii)$

$\begin{aligned} &\therefore AC \parallel XX' \\ &\therefore CO = AB \end{aligned}$

From equation (i) and (ii)

$$\frac{OB}{OB'} = \frac{OF_1}{B'F_1} \Rightarrow \frac{OB}{OB'} = \frac{OF_1}{OF_1 - OB'}$$

By using sign convention,

OB = distance of object = -u  
 OB' = distance of image = -v  
 OF<sub>1</sub> = focal length = -f

$$\begin{aligned} \therefore \frac{OB}{OB'} &= \frac{OF_1}{OF_1 - OB'} = \frac{-u}{-v} = \frac{-f}{-f - (-v)} \\ \Rightarrow \frac{u}{v} &= \frac{-f}{-(f-v)} \\ \Rightarrow u(f-v) &= vf \\ \Rightarrow uf - uv &= vf \end{aligned}$$

Both sides divided by uvf

$$\begin{aligned} \frac{uf}{uvf} - \frac{uv}{uvf} &= \frac{vf}{uvf} \\ \Rightarrow \frac{1}{v} - \frac{1}{f} &= \frac{1}{u} \end{aligned}$$

$\Rightarrow \frac{1}{f} = \frac{1}{v} - \frac{1}{u}$

➔ Which is required lens formula

- **Define magnification and obtain the expression for magnification**
- The ratio of height of images & the height of object with their sign is called magnification.

$\therefore \text{magnification} = \frac{\text{Height of image}}{\text{Height of object}}$

- Magnification is denoted by 'M'.

Let,

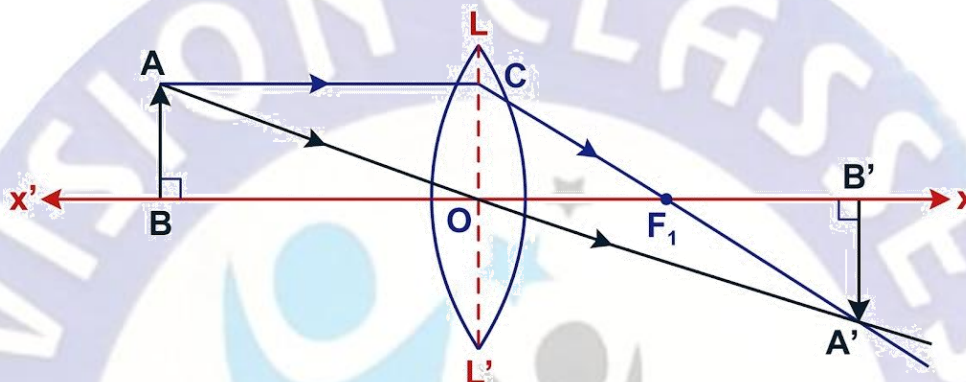
The height of image =  $h_i$

The height of object =  $h_o$

$$\therefore m = \frac{h_i}{h_o} \quad \dots\dots\dots (i)$$

• **For magnification formula**

- Let LL' is a convex lens which has optical centre 'O', centre of curvature 'C', radius of curvature 'R', principal focus 'F', focal length 'f' and principal axis XX'.
- A luminous object AB is placed at XX'. An image A'B' is formed by convex lens.



In  $\triangle AOB$  and  $\triangle A'OB'$   
 $\angle AOB = \angle A'OB'$  [V.O angle]  
 $\angle ABO = \angle A'B'O = 90^\circ$

$\therefore$  by A – A similarity  
 $\triangle AOB \sim \triangle A'OB'$

$$\therefore \frac{AB}{A'B'} = \frac{OB}{OB'} \quad \dots\dots\dots (ii)$$

By using sign convention,

AB = height of object =  $h_o$   
 $A'B'$  = height of image =  $-h_i$   
 OB = distance of object =  $-u$   
 $OB'$  = distance of image =  $v$

From equation (ii)

$$\frac{h_o}{-h_i} = \frac{-u}{v} \Rightarrow \frac{-h_i}{h_o} = \frac{-v}{u}$$

$$\Rightarrow \frac{h_i}{h_o} = \frac{v}{u} \quad \Rightarrow m = \frac{v}{u} \quad \text{[from equation (i) } \frac{h_i}{h_o} = m \text{ ]}$$

$\therefore$  Required formula for magnification (M)

$$\therefore m = \frac{v}{u}$$

- **Relation between m, v and f**

∴ lens formula:  $\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$

Both sides multiplied by v

$$\frac{v}{f} = \frac{v}{v} - \frac{v}{u} \Rightarrow \frac{v}{f} = 1 - \frac{v}{u}$$

$$\Rightarrow \frac{v}{f} = 1 - m \quad \Rightarrow m = 1 - \frac{v}{f}$$

$$\Rightarrow m = \frac{f-v}{f}$$

→ Which is required relationship.

**Note:**

1.  $m = -ve$   
Image is real & Inverted
2.  $m = +ve$   
Image is virtual & erect

**Note:**

1. If  $m = +ve$ , i.e, the image is virtual and erect.
2. If  $m = -ve$ , i.e, the image is real and inverted.

- **Relative refractive index**

- When light is going from one medium (other than vacuum or air) to another medium then the value of refractive index is called relative refractive index.

For example:-

- ✓ The refractive index of glass with respect to water is 1.125.

- **Absolute refractive index**

- When light is going from vacuum or air to another medium then the value of refractive index is called absolute refractive index.

OR

The refractive index of a medium with respect to air is called its absolute refractive index.

For example:-

- ✓ Refractive index of water with respect to air is 1.33.